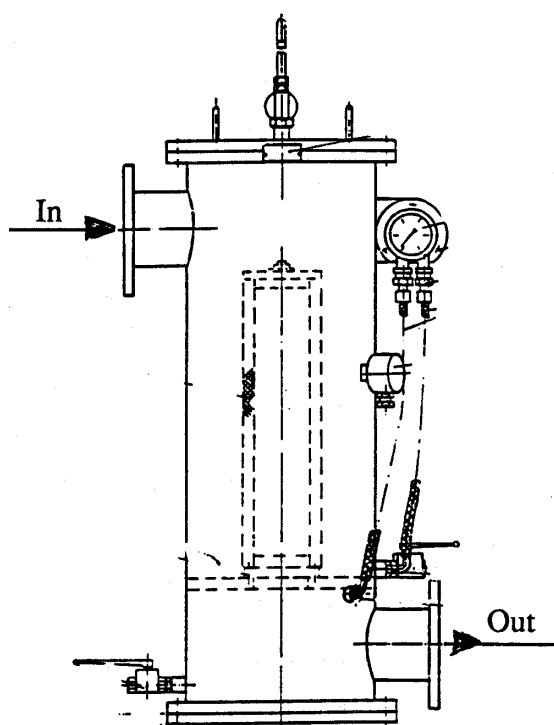


Service Bulletin

MAKE YOU UP TO DATE WITH THE LATEST NEWS FROM FRANK MOHN SERVICES

Oil cleanliness - Filtration

Main key to achieve a long lasting well functioning cargo pumping plant-
and improvement of the total economy



- ★ Oil sampling every 3rd. month
- ★ Particle counting ISO 4406
- ★ Never mix the hydr. oil with other brands of lub. oil.
- ★ Pressure drop over the filter combined with particle counting is the only safe indicator when to change the filterelements.
- ★ New filters with bigger dirt-capacity give longer service intervals and improved economy
- ★ Cheap- non original- filters are always costly in the end.



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DEAR CUSTOMER!

In this Service Bulletin we will highlight the hydraulic oil and the filtration.

As we all know the cleanliness of the oil as well as the general oil condition is very important for the lifetime of the system.

The practical life has shown that some of the "FRAMO ships" are sailing around with far too high particle level in the hydr. oil.

Our recommended level of particles in the system is:

ISO 4406 code 16/12.

This means:

Number of particles per ml $\geq 5 \mu\text{m}$: 320, max 640

Number of particles per ml $\geq 15 \mu\text{m}$: 20, max 40

With FRAMO original filter element this cleanliness level should not be of any problem to maintain.

Nevertheless, we have found ships sailing around with more than 2-3 thousand particles per ml between 5 - 15 μm .

This is very costly in the end, and we therefore strongly advise you to pay attention to the filtration and the cleanliness of the hydr. oil.

As you know the hydr. oil has four main functions in the system:

- 1: Transmit the power from the aggregates to the different consumers.
- 2: Lubrication of all moving parts.
- 3: Transfer all produced heat back to the cooler.
- 4: Transfer all dirt back to the filter.

Do not play around with non FRAMO oil filter elements.

Be aware of the following important filter specifications:

1) **The filter Beta Ratio (β)**

Normal FRAMO filters have a Beta Ratio = $\beta_{12} \geq 75$.

This means if 75 particles of 12 μm or bigger come into the filter only one passes through. In other words the Beta Ratio is the filter's ability to stop the particles, i.e.

$$\beta = \frac{\text{particles in}}{\text{particles out}}$$

2) **The filter dirt capacity**

Filter with higher dirt capacity gives longer service intervals and improved economy.

Particles counting combined with pressure drop over the filter is the only safe indicator when to change the filter element.

FRAMO offices in Bergen, Rotterdam, Houston and Singapore can offer hydr. oil monitoring and particle counting.

It is essential that the oil sampling is provided regularly, from the same highlighted sampling point, and in an absolutely clean sampling bottle.



FRAMO systems designed before June 1983 are using ISO VG32 oil, and systems designed after June 1983 are using ISO VG46 oil.

For the VG46 oil, FRAMO filter element ID no. 109637, was replaced in 1992 by ID no. 3025780 with approx. 3 times bigger dirt capacity than the old type. The new element fits into the old filter box.

In addition to the particle counting the oil analysis shall as a minimum give the following information:

<u>TEST</u>	<u>TEST METHOD</u>
Specific gravity.....	ASTM D 1298
Flash point.....	ASTM D 92
Viscosity cSt at 40°C.....	ASTM D 445
Viscosity cSt at 100°C.....	ASTM D 445
TAN (total acid No.) mg KOH/g	ASTM D 664
Water %.....	ASTM D 95
Zinc	PPM
Phosphore	PPM
Chlorides	PPM
Calcium	PPM

General acceptance criteria can not be given as they will vary for different oils, but the following can be said:

Water.....	Max 0,05 or 500 ppm
Zinc.....	Max reduction 25% of original value
Phosphore.....	Max reduction 25% of original value
Chlorides.....	
Calcium.....	Depends on type of oil
Viscosity 40°C VG46 oil.....	Min 38 cSt
Viscosity 100°C VG46 oil.....	Min 6,2 cSt

For the rest of the analyse, the oil company will inform if the oil is fit for further use.

Factors affecting the lifetime of the oil are:

Specific gravity: A significant change indicates mixing with another liquid.

Flash point: A significant change indicates mixing with another liquid.

Viscosity: Reduced viscosity indicates:

- Insufficient shear stability
- Mixing with another liquid

Increased viscosity indicates:

- Excessive oxidation
- Mixing with another liquid

If the viscosity drops below 38 cSt at 40°C or 6,2 cSt at 100°C, the oil does no longer meet the ISO VG46 grade.
If the rest of the analysis confirm that the oil is in a good condition, replenish with ISO VG68 oil of the same type as in the system.



- Tan:** Critical values depend on the oil used and must be informed by the oil supplier.
- Water:** High water content will:
- Increase wear in the system
 - Increase corrosion in the system
 - Damage the additive package causing clogged filters for some types of hydraulic oils. If the water exceeds 0,05%, it has to be removed. Centrifugal separators are not permitted used as it can damage the additive package. Contact Frank Mohn Services AS for help.
- Chlorides:** Chlorides have the same effect as water. In addition it will accelerate corrosion in the system.
- Calcium:** A certain amount of calcium is a natural part of the additive package in some hydraulic oils. An increase in the calcium level indicates mixing with another type of oil (for ex. motor oil). Calcium added to the oil this way can damage the normal additive package of the hydraulic oil causing a total breakdown of the system.
- Zinc and Phosphre:** This is for most hydraulic oils the normal antiwear additive for the oil and should be monitored carefully. A decreased level will cause increased wear in the system. At a decrease of 25% of the values for new oil, counter measures such as:
- Replenish with normal fresh oil or
 - Replenish with new oil specially blended with a higher level of antiwear additives should be taken.

If any further information is needed, please contact FRAMO Service offices world-wide!